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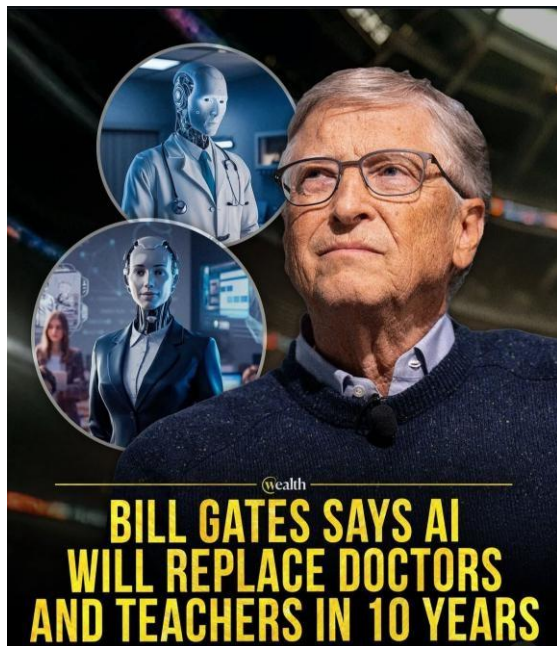
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Beyond the Counter: How AI Is Rewriting the Pharmacy Playbook

What if your next prescription was generated by a machine instead of your pharmacist?

Artificial intelligence (AI) is rapidly changing the way healthcare works — and pharmacy is no exception. From speeding up drug discovery to helping manage medications more safely and effectively, AI is opening the door to faster, more accurate, and personalised care. Smarter algorithms and powerful data tools can now predict how drugs will work^{12,13}, identify new treatment options^{14,15}, and support pharmacists in creating medication plans tailored to individual patients.^{16,17} In everyday pharmacy settings, AI can automate time-consuming tasks like reading prescriptions and checking for drug interactions, while pharmacy apps powered by AI are offering round-the-clock support to users at home — especially those in remote or hard-to-reach areas. Despite its promise, AI isn't without challenges. It still can't replace the empathy and human connection that pharmacists offer, and its effectiveness depends on the quality and fairness of the data it uses. There are also concerns around privacy, ethics, and the need for ongoing technical support. This research explores how AI can work alongside pharmacists — not replace them — by focusing on areas like patient education, medication adherence, and real-life implementation.

AI first entered healthcare through data-heavy fields like medical imaging and diagnostics, where it helped identify patterns and predict risks.¹ In the pharmaceutical industry, AI was adopted for drug discovery, accelerating the identification of new compounds and streamlining development. As its effectiveness grows, AI continues to expand into clinical decision support, personalised medicine, and pharmacy operations.¹



(wealth on Instagram, 2020)

The financial commitment to AI in the pharmaceutical sector underscores its transformative potential. The global AI in drug discovery market was valued at approximately USD 1.5 billion in 2023² and is projected to expand at a compound annual growth rate (CAGR) of 29.7%,² reaching USD 9.1 billion by 2030.² Similarly, Fortune Business Insights reported a market valuation of USD 3.54 billion in 2023,³ with expectations to grow to USD 7.94 billion by 2030,³ reflecting a CAGR of 12.2%.³ This substantial growth is driven by the increasing adoption of AI technologies aimed at enhancing efficiency and reducing costs in drug discovery and development.

Healthcare spending accounts for 6–7% of global GDP (about \$8.5 to \$9 trillion annually),⁴ and developing a new drug can cost over \$1 billion and take up to 14 years,⁴ with very low success rates—particularly in cancer, where 97% of drugs fail during clinical trials.⁴ This makes drug development risky and drives up drug prices. The digitisation of biomedical data and advances in AI, particularly deep learning and graph-based models, are transforming the drug discovery pipeline — from target identification and molecular design to clinical trials and FDA (U.S. Food and Drug Administration) approval.

AI significantly enhances pharmacovigilance (Pharmacovigilance is all about monitoring the safety of medicines after they are approved and used by the public. It helps identify unexpected side effects, drug interactions, or long-term risks that may not have been discovered during clinical trials) by analysing real-world data to predict and detect adverse drug events (ADEs).⁷ Using machine learning (ML)⁷ and natural language processing (NLP)⁷, AI automates safety report processing, improves consistency, reduces bias, and streamlines post-marketing surveillance activities. A study by Fan et al. showed that deep learning models like BERT could effectively detect unreported side effects from online drug reviews.⁵

AI also supports personalised medicine by predicting drug responses and identifying patient-specific molecular patterns. Integrating genomic data with AI helps tailor drug selection, avoiding ineffective or harmful treatments. This approach is especially crucial in conditions like epilepsy, where many patients do not respond to standard medications, leading to reduced quality of life and higher healthcare costs.

"If we do this right, AI can restore the care in healthcare — by reducing the burden of documentation, improving diagnosis, and giving clinicians more time to connect with patients."⁶

— Dr. Eric Topol, Deep Medicine (2019)

This quote perfectly aligns with the hybrid model we are exploring here in this project — where AI supports clinicians rather than replacing them — and reinforces my message about maintaining empathy and human connection in healthcare.

Ballad Health has partnered with MedAware to implement an AI-driven medication safety monitoring platform aimed at enhancing patient safety and optimizing pharmacy workflows.⁸ This system integrates with Ballad Health's electronic health record (EHR) to provide real-time, patient-specific analysis of medication-related risks. By continuously evaluating prescribed drugs against up-to-date patient profiles, the platform identifies potential medication errors, allowing healthcare providers to intervene promptly and prevent adverse drug events. The AI utilizes advanced machine learning algorithms to analyse prescribing patterns and detect outliers that may indicate harmful or inappropriate prescriptions. This proactive approach has led to significant improvements in patient safety, with Ballad Health's pharmacy teams taking action on over 70% of the risks identified by the system.⁸

Additionally, the integration of MedAware's platform has reduced alert fatigue among clinicians by providing more accurate and actionable alerts, thereby enhancing the overall efficiency of the pharmacy operations.⁸ This collaboration exemplifies how AI can be effectively utilized to support healthcare professionals in delivering safer and more efficient patient care.



OpenAI. (2025)

Some AI tools in healthcare are not considered clinical decision support systems because they operate independently of human input and do not assist in real-time decision-making. For example, robotic surgery systems like the da Vinci robot can perform precise procedures, but they follow programmed instructions and don't help decide if or how a patient should be treated.⁹ Imaging AI may highlight abnormalities on scans, but if it's only improving image quality without influencing diagnosis or treatment plans, it acts more as a background tool.¹⁰ Similarly, automated devices like insulin pumps deliver medication based on preset rules without clinician involvement¹¹ — they replace decision-making rather than support it. Lastly, administrative AI, such as tools that transcribe clinical notes or manage patient records, helps streamline workflows but doesn't guide clinical care. These tools are valuable, but they don't directly contribute to a clinician's judgment or treatment decisions, which is why they are excluded from definitions of clinical decision support in research.

While artificial intelligence offers remarkable speed and precision, it lacks the emotional intelligence, contextual understanding, and ethical sensitivity that human pharmacists bring to patient care. Pharmacists don't just dispense medications — they listen, reassure, and interpret non-verbal cues that AI systems simply cannot perceive. In moments of fear or uncertainty, a calm explanation or a kind word from a trusted pharmacist can make a world of difference. This is where the motif of balance becomes essential. If AI is the engine that drives efficiency, then the human is the compass — providing ethical direction and emotional grounding. Without that, technology risks becoming a “compass without a true north” — full of information, but lacking wisdom.

Therefore, we need hybrid AI-human models. In hybrid models, AI doesn't replace pharmacists — it liberates them. By taking over repetitive, data-driven tasks, AI gives pharmacists the freedom to focus on what machines cannot do: build trust, provide emotional support, and make ethically nuanced decisions. The future of pharmacy is not one of replacement, but of collaboration between intelligence and empathy, between algorithms and human connection.

THE FUTURE: A VISION FOR HYBRID PHARMACY SYSTEMS

Workflow-Integrated AI Support

In this model, AI tools are embedded directly into pharmacists' workflows to assist with routine, data-heavy tasks:

AI handles: Prescription screening, drug interaction alerts, dosage checks, real-time adherence tracking.

Pharmacist handles: Interpretation of AI output, patient counselling, resolving ambiguities, ethical decision-making.

1



2



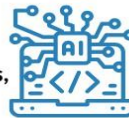
AI-Assisted Clinical Decision Support

This model uses AI for real-time decision support, especially in hospital and clinical pharmacy settings:

AI provides: Patient-specific treatment suggestions based on EHR data, lab results, and drug history.

Pharmacist evaluates: Clinical relevance, feasibility, and patient preferences before applying recommendations.

3



AI-Driven Patient Engagement Platforms

Here, AI tools are used to extend pharmacist support outside the pharmacy, improving long-term care and communication:

AI manages: Refill reminders, medication tracking apps, symptom checkers, chatbot-based Q&A for common concerns.

Pharmacist steps in: For escalated concerns, emotional support, or detailed follow-ups.

4



AI-Enhanced Pharmacogenomics & Personalised Medicine

This forward-looking model integrates AI with genomic data to guide precise drug selection:

AI analyses: Genetic profiles to predict drug response, recommend tailored therapies, or flag high-risk medications.

Pharmacist collaborates: With prescribers and patients to interpret the data in the context of the patient's values, history, and lifestyle.



The future of pharmacy lies not in choosing between human expertise and artificial intelligence, but in combining their strengths. Hybrid AI-human models offer a path forward—where technology supports pharmacists in delivering safer, more efficient, and more personalised care. With thoughtful implementation, ethical oversight, and a focus on collaboration, we can ensure AI enhances rather than replaces the human touch at the heart of healthcare.

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